

HAT-P-12b

R-0748

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_240414 · created 2026-07-28T02:40:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460414.7614 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.01
Transit depth (Rp/Rs)^2	1.71 +/- 0.27 [%]
Orbital Inclination (inc)	88.04 +/- 0.77
Airmass coefficient 1 (a1)	0.9081 +/- 0.0096
Airmass coefficient 2 (a2)	0.09 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	#3 - [153, 359]
Optimal Aperture	4.06
Optimal Annulus	9.46
Transit Duration (day)	0.0907 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.01 vs 0.1406 (deviation 0.96σ)
Duration fit vs published	0.0907 d vs 0.0975 d (deviation 1.17σ)
Mid-time O–C	-1.1 min
Depth SNR	13.1
Residual scatter	1.06 %

Light curve

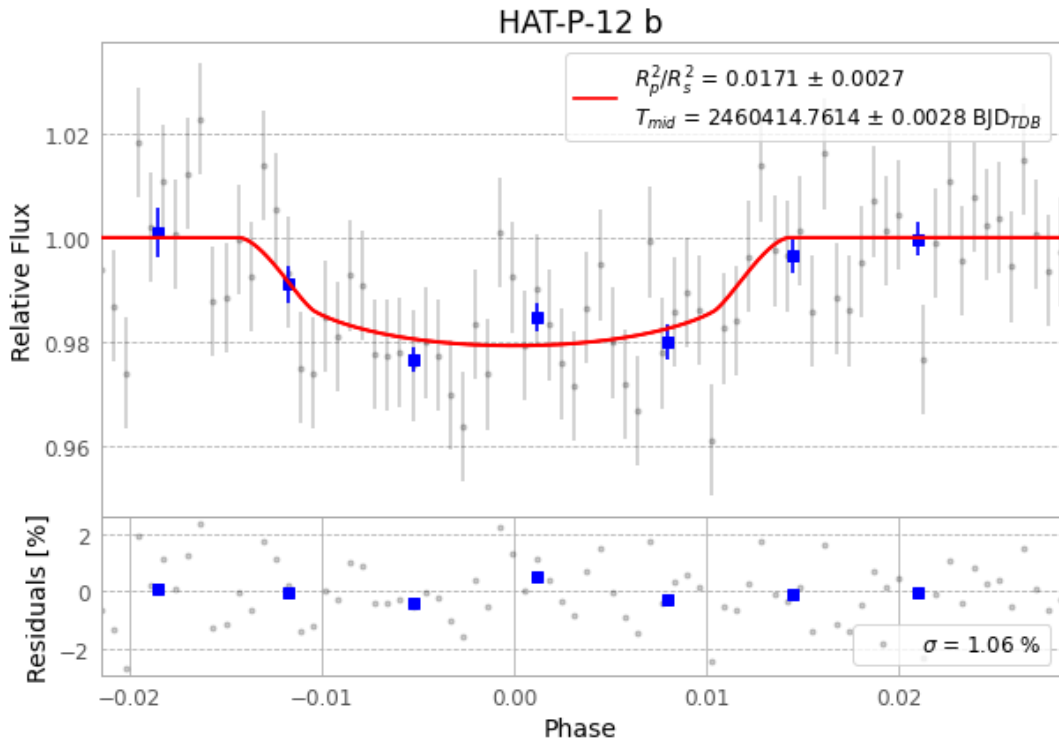


Figure 1 — FinalLightCurve_HAT-P-12 b_2024-04-13.png (SHA-256 fb1e75c98826735e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a04ccc72351a87f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

78 FITS frames (51 MB), HATP-12240414043015.FITS → HATP-12240414082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-240414.log (0e3e79d0e830...), FinalLightCurve_HAT-P-12 b_2024-04-13.png (fb1e75c98826...), FinalLightCurve_HAT-P-12 b_2024-04-13.csv (4b733e42e309...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:40Z.